

Recall from last week:

- A **linear equation** in the variable $x_1, \dots, x_n$ is an equation that can be put in the following form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where b and the coefficients $a_1, \dots, a_n$ are real or complex numbers. A **system of linear equations** is a collection of one or more linear equations involving the same variables. A **solution** to the linear system is a list of $(s_1, \dots, s_n)$ of numbers that are solutions to each of the linear equations in the system.

- A linear system is **consistent** if there is a solution and **inconsistent** if there is no solution.
- A linear system has either no solution, exactly one solution, or infinitely many solutions.
- The following elementary row operations preserve solutions.
 1. Replace one row by the sum of itself and a multiple of another row.
 2. Interchange two rows.
 3. Multiply all entries in a row by a non-zero constant.
- Two matrices are called **row equivalent** if there is a sequence of elementary row operations changing one matrix to another.
- If the augmented matrices of two linear systems are row equivalent, then they both have the same solution set.
- Proof Techniques
 1. Quantifiers and counterexamples.
 2. Direct Proofs.
 3. Proof by cases.
 4. Proof by contrapositive.

Definition 1. *A matrix is in row echelon form if:*

1. *All rows consisting of only zeros are at the bottom.*
2. *The leading coefficient (also called the pivot) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.*

1.

(a) Give an example of a matrix in echelon form.

(b) Give an example of a matrix not in echelon form.

Definition 2. *A matrix in row echelon form is in reduced row echelon form if*

- (a) *The leading entries are all 1.*
- (b) *Each leading 1 is the only non-zero entry in its column.*

(c) Give an example in reduced row echelon form.

Fact: Each matrix is row equivalent to one and only one reduced echelon matrix.

Definition 3. A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced row echelon form. A **pivot column** is a column that has a pivot position.

Row Reduction Algorithm (Gaussian Elimination).

- (a) The leftmost non-zero column is a pivot column.
 - (b) Select a non-zero entry in the pivot column as pivot. Interchange rows to move this entry to the top.
 - (c) Use elementary row operations to create zeros below the pivot.
 - (d) Apply (a)-(c) to the submatrix obtained by removing pivot row and column. Repeat until all non-zero rows are modified.
 - (e) Beginning with rightmost pivot and working upwards and to the left, create 0's above each pivot. Scale pivot positions entries to be 1.
2. Row-reduce the following matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

3. Use row-reduction to solve the linear system

$$\begin{cases} x_1 + 3x_2 + 4x_3 = 7 \\ 3x_1 + 9x_2 + 7x_3 = 6 \end{cases}$$

Definition 4. *The variables corresponding to the pivot columns in an augmented matrix are called **basic** and the others are called **free**.*

Facts: A linear system is consistent if and only if the rightmost column in the augmented matrix is not a pivot column. If the system is consistent, then it has a unique solution when there are no free variables and infinite solutions when there is one or more free variables.

4. Find the general solution of the augmented matrix :

$$\left(\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array} \right)$$

Definition 5. A matrix with 1 column is called a **column vector**. A matrix with one row is called a **row vector**.

We can add vectors of the same size by adding the corresponding components; namely, if $\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then $\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$. Similarly, we can scale a vector by a real

number c , by multiplying all the components by c ; namely, $c\vec{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$.

Definition 6. Given $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$ and scalars $c_1, \dots, c_p$, the vector

$$\vec{y} = c_1\vec{v}_1 + \dots + c_p\vec{v}_p$$

is called the **linear combination** of $\vec{v}_1, \dots, \vec{v}_p$ with weights $c_1, \dots, c_p$.

5. Suppose $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$; and $\vec{w} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. Can $\vec{w}$ be expressed as a linear combination of $\vec{u}$ and $\vec{v}$?

6. Suppose $\vec{a} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$; $\vec{b} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$; $\vec{c} = \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix}$; and $\vec{d} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Can $\vec{d}$ be expressed as a linear combination of $\vec{a}$, $\vec{b}$, and $\vec{c}$?

A vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

has the same solutions as the augment matrix

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \left| \vec{b} \right].$$

Hence $\vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ if and only if there is a solution to this matrix.

Definition 7. If $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \dots, \vec{v}_p$ is called the **span** of $\vec{v}_1, \dots, \vec{v}_p$. We write this set as $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$.

7.

(a) Suppose $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$; and $\vec{b} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Is $\vec{b}$ in $\text{Span}\{\vec{u}, \vec{v}\}$?

(b) Does there exist $\vec{b}$ not in $\text{Span}\{\vec{u}, \vec{v}\}$?

Row Echelon Form and Vector Equations – Answers / Solutions

1. (There are many examples for each of these.)

(a)

$$\begin{bmatrix} 0 & 1 & 10 & \pi \\ 0 & 0 & 0 & \ln(2) \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 0 & 1 & 10 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Follow the algorithm:

The left column is nonzero, so it is a pivot column. The top entry 1 is not zero, so we use this for our pivot. Then we create zeroes below the pivot:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{R_2 + (-2)R_1 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{R_3 + (-3)R_1 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix}$$

Now we focus on the submatrix

$$\begin{bmatrix} 0 & -2 & -8 \\ 0 & -3 & -12 \end{bmatrix}$$

The leftmost nonzero column is the second column, so it is a pivot column. The top entry -2 is not zero, so we use this for our pivot. Then we create zeroes below the pivot:

$$\begin{bmatrix} 0 & -2 & -8 \\ 0 & -3 & -12 \end{bmatrix} \xrightarrow{R_2 + (-\frac{3}{2})R_1 \rightarrow R_2} \begin{bmatrix} 0 & -2 & -8 \\ 0 & 0 & 0 \end{bmatrix}$$

Putting this back into the original matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 + 2R_2 \rightarrow R_1} \begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(-\frac{1}{2})R_2 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. Write this as an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array} \right]$$

and follow the algorithm again. The left column is nonzero, so it is a pivot column. The top entry 1 is not zero, so we use this for our pivot. Then we create zeroes below the pivot.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array} \right] \xrightarrow{R_2 + (-3)R_1 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array} \right]$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array} \right] \xrightarrow{R_1 + \frac{4}{5}R_2 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & -5 & -15 \end{array} \right] \xrightarrow{\frac{-1}{5}R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Translating back to equations:

$$\begin{cases} x_1 + 3x_2 = -5 \\ x_3 = 3 \end{cases}$$

So there are infinitely many solutions.

4. Follow the algorithm again. The left column is nonzero, so it is a pivot column. The top entry 3 is not zero, so we use this for our pivot. Then we create zeroes below the pivot.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array} \right] \xrightarrow{R_2 + (-3)R_1 \rightarrow R_2} \left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & -4 & 8 & 0 \end{array} \right] \xrightarrow{R_3 + (-2)R_1 \rightarrow R_3} \left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\frac{1}{3}R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So this system is consistent with one basic variable (x_1) and two free variables (x_2, x_3). General solutions to this system have the form:

$$x_2 \begin{bmatrix} \frac{2}{3} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

5. No. Since $\vec{v} = 2\vec{u}$, then for any scalars c_u, c_v , the entries of $c_u\vec{u} + c_v\vec{v}$ are equal:

$$c_u\vec{u} + c_v\vec{v} = c_u\vec{u} + 2c_v\vec{u} = (c_u + 2c_v)\vec{u}$$

6. We could try the same approach as the previous problem, but when the vectors get larger and more complicated, it can be difficult to analyze them all. Let's use a more systematic approach, and change this to a row reduction problem.

We are asking whether there are scalars c_a, c_b, c_c such that $c_a\vec{a} + c_b\vec{b} + c_c\vec{c} = \vec{d}$. This is a system of linear equations with the variables c_a, c_b, c_c :

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -1 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array} \right]$$

Now we follow the row reduction algorithm. The leftmost column is nonzero, and the top entry 1 is nonzero, so this is our pivot.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -1 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array} \right] \xrightarrow{R_2+R_1 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & 8 & 6 \end{array} \right]$$

Now the second column is nonzero and the leading entry of the second row is 1, so this is our second pivot.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & 8 & 6 \end{array} \right] \xrightarrow{R_3+(-2)R_2 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 10 & 4 \end{array} \right]$$

Now we create zeroes above pivot positions, and scale the pivot positions to be 1.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 10 & 4 \end{array} \right] &\xrightarrow{\frac{1}{10}R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & \frac{2}{5} \end{array} \right] \xrightarrow{R_2+R_3 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 0 & \frac{7}{5} \\ 0 & 0 & 1 & \frac{2}{5} \end{array} \right] \\ &\xrightarrow{R_1+(-5)R_3 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{7}{5} \\ 0 & 0 & 1 & \frac{2}{5} \end{array} \right] \end{aligned}$$

So we have our solution $c_a = 0, c_b = \frac{7}{5}, c_c = \frac{2}{5}$. Just to be safe, let's check that this actually works:

$$0 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \frac{7}{5} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \frac{2}{5} \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} = \vec{d}$$

7. (a) Row reduce the augmented matrix:

$$\begin{aligned} \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 2 & 4 & 4 \end{array} \right] &\xrightarrow{R_2+(-2)R_1 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & -2 & 6 \end{array} \right] \xrightarrow{-\frac{1}{2}R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & 1 & -3 \end{array} \right] \\ &\xrightarrow{R_1+(-3)R_2 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & 8 \\ 0 & 1 & -3 \end{array} \right] \end{aligned}$$

so $\vec{b} \in \text{span}\{\vec{u}, \vec{v}\}$ (Check: $8 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-3) \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \vec{b}$).

(b) If $\vec{b} \notin \mathbb{R}^2$, then $\vec{b}$ is not in the span of $\vec{u}$ and $\vec{v}$. This is a somewhat silly answer to this question. Now assume $\vec{b} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. Then we row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 3 & x \\ 2 & 4 & y \end{array} \right] \xrightarrow{R_2 + (-2)R_1 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & x \\ 0 & -2 & y - 2x \end{array} \right] \xrightarrow{-\frac{1}{2}R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & x \\ 0 & 1 & \frac{2x-y}{2} \end{array} \right]$$

$$\xrightarrow{R_1 + (-3)R_2 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & -2x + \frac{3}{2}y \\ 0 & 1 & \frac{2x-y}{2} \end{array} \right]$$

so $\vec{b} \in \text{span}\{\vec{u}, \vec{v}\}$. (Check $(-2x + \frac{3}{2}y) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2x-y}{2} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \vec{b}$.)